Energy decay and iterate nonconvergence for Lyapunov damping

Abstract

We prove a full-energy bound of order $\log^4(k)/k^2$ for the normalized Lyapunov damping gradient iteration on smooth convex functions with an attained minimum, without assuming boundedness of the orbit or sublevels. The estimate holds for every fixed step $0 < s \leq 1/L$, including equality, and has an explicit initial-data constant. A bounded initial sublevel improves the bound to $O(k^{-2})$. A fixed scalar convex function with Lipschitz gradient and bounded sublevels has an equal-start orbit whose cluster set is an entire minimizer interval; on that same orbit, both energy and objective fail to be $o(k^{-2})$. For the unnormalized continuous law, every given global classical solution satisfies an explicit $O(\log^4(t)/t^2)$ energy bound without a radius assumption, while bounded solutions satisfy $O(t^{-2})$. A scalar quadratic has $t^2E(t) \to 4$ and objective scaled limsup 4 and liminf 0. The unbounded-orbit endpoints without logarithmic loss are not established.

Keywords: convex optimization, Lyapunov damping, inertial gradient method, energy decay

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1 Introduction

Lyapunov damping chooses the friction coefficient from the current mechanical energy. Maier, Castera, and Ochs introduced this principle and its normalized gradient discretization, LYDIA [1, 2]. The damping vanishes with the energy and does not require a prescribed iteration-dependent schedule. Both formulations use the minimum value of the objective.

In the publicly available version [1, Theorem 5.1], the discrete energy is nonincreasing, and its convergence to zero follows in finite dimensions when the trajectory has a bounded subsequence. Remark 5.2 there leaves the discrete convergence rate for further work. We give an explicit $O(k^{-2})$ estimate for the full energy on bounded initial sublevels, and an initial-data estimate $O(\log^4(k)/k^2)$ without any boundedness assumption. The proof uses the standard convex gradient comparison underlying energy monotonicity, followed by a positive correction involving distance to a minimizer. For the bound without a radius, the damping-drop term controls the growth of a normalized correction, and a count of the iterations spent at each energy level extracts the rate.

The discrete energy rate does not force point convergence. The flat-bottom quadratic family already appears in the numerical experiments of [1, Section 5.2 and Figure 5]. For one fixed member, a strict step and a prescribed equal start, we prove that the exact discrete orbit has the entire minimizer interval as its cluster set. The same orbit has

positive limsups for both k^2E_k and $k^2(f(x_k) - f_\star)$, showing that the $O(k^{-2})$ energy and objective orders under bounded sublevels are sharp. The argument uses actual boundary crossings and two exact exterior updates.

The continuous result in [1, Theorem 1.1] gives objective decay $o(t^{-2+\delta})$ for every $\delta > 0$ along bounded trajectories. Its concluding section asks about the endpoint $\delta = 0$. For bounded classical solutions, we obtain a full-energy bound $O(t^{-2})$. For arbitrary given global classical solutions, we prove $O(\log^4(t)/t^2)$ without a radius assumption. This retains $o(t^{-2+\delta})$ for every $\delta > 0$ in the absence of the boundedness hypothesis discussed in the conclusion of that public version. We then compute the exact energy asymptotic and the objective oscillation for one scalar quadratic. It excludes the literal little-o endpoint for that continuous law. The distinction between energy and objective is essential: the energy has a positive scaled limit, while the objective repeatedly vanishes.

The comparisons in this note refer specifically to arXiv:2311.10053v2, dated 15 April 2024 [1]. The final journal bibliographic record is [2]; its main discrete theorem and concluding discussion were not available for a complete version comparison. Accordingly, no assertion is made that these questions remain open in the final journal article. The statements and proofs below concern the explicitly displayed iteration and differential equation.

2 The normalized discrete iteration

Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and continuously differentiable, with L-Lipschitz continuous gradient, where $L > 0$. Suppose that

$$S = \operatorname{argmin} f \neq \varnothing, \qquad f_\star = \min f.$$

Choose $x_{-1} = x_0$ with $E_0 = f(x_0) - f_\star > 0$, and a fixed step $0 < s \leq 1/L$. For each integer $k \geq 0$, define

$$\begin{aligned} v_k &= x_k - x_{k-1}, & E_k &= f(x_k) - f_\star + \frac{\|v_k\|^2}{2s}, \\ Q_k &= \sqrt{E_k}, & a_k &= \frac{Q_k}{Q_0}, \\ y_k &= x_k + (1 - a_k)v_k, & x_{k+1} &= y_k - s\nabla f(y_k). \end{aligned} \tag{1}$$

In particular, $a_0 = 1$ and $x_1 = x_0 - s\nabla f(x_0)$. This explicitly fixes the initial update; the proof of [1, Theorem 5.1] also begins at $k = 0$. The coefficient a_k here is the normalized damping coefficient denoted by β_k in that source.

Theorem 1. *Suppose that the initial sublevel*

$$\mathcal{L}_0 = \{x \in \mathbb{R}^d : f(x) \leq f(x_0)\}$$

is bounded. Fix any $z \in S$, and put

$$R = \sup_{x \in \mathcal{L}_0} \|x - z\|, \qquad C = \frac{R^2}{2sQ_0} + R\sqrt{\frac{2}{s}} + 7Q_0. \tag{2}$$

Then every integer $k \geq 2$ satisfies

$$E_k \leq \frac{4C^2}{k^2}, \qquad f(x_k) - f_\star \leq \frac{4C^2}{k^2}, \qquad \|x_k - x_{k-1}\| \leq \frac{2\sqrt{2s}\,C}{k}. \tag{3}$$

The radius in (2) is the radius of the whole initial sublevel relative to the chosen minimizer, rather than only the initial distance. The theorem does not give point convergence, a distance-to-solution rate, or a claim for unbounded initial sublevels. No optimality is asserted for the constant. If $f(x_0) = f_\star$, the equal-start iteration is taken to be stationary; the formula for C is only used when $Q_0 > 0$.

2.1 The gradient comparison and monotone energy

For arbitrary $y, u \in \mathbb{R}^d$ and $x^+ = y - s\nabla f(y)$, convexity and smoothness give

$$f(x^+) \leq f(u) + \frac{\|y - u\|^2 - \|x^+ - u\|^2}{2s} - \frac{1 - sL}{2s} \left\|x^+ - y\right\|^2. \tag{4}$$

Indeed, the usual smoothness bound at y, followed by the convexity inequality with comparison point u, gives

$$f(x^+) \leq f(u) + \left\langle \nabla f(y), x^+ - u \right\rangle + \frac{L}{2} \left\|x^+ - y\right\|^2.$$

Substitute $s\nabla f(y) = y - x^+$ and expand the inner product to obtain (4). Its last term is nonpositive throughout the stated step range.

Write

$$F_k = f(x_k) - f_\star, \quad K_k = \frac{\|v_k\|^2}{2s}, \quad h_k = \frac{\|x_k - z\|^2}{2s}, \quad p_k = \frac{\langle x_k - z, v_k \rangle}{s}. \tag{5}$$

Applying (4) with $u = x_k$ and adding K_{k+1} yields

$$E_{k+1} \leq F_k + (1 - a_k)^2 K_k = E_k - a_k(2 - a_k)K_k. \tag{6}$$

Starting from $a_0 = 1$, induction proves

$$0 \leq E_{k+1} \leq E_k \leq E_0, \qquad 0 \leq a_{k+1} \leq a_k \leq 1. \tag{7}$$

In particular, $x_k \in \mathcal{L}_0$ and $\|x_k - z\| \leq R$ for all k. If $E_j = 0$ at a finite index, then $x_j \in S$ and $v_j = 0$. The recurrence gives $x_k = x_j$ and $E_k = 0$ for every $k \geq j$. All divisions by an energy below are restricted to its positive range.

2.2 A positive auxiliary energy

Define

$$\begin{aligned} G_k &= (1-a_k)p_k + a_k h_k - (2-a_k)K_k, \\ \Phi_k &= G_k + 3Q_0 Q_k, \qquad \Psi_k = \Phi_k + E_k. \end{aligned} \tag{8}$$

These expressions are defined even at a zero-energy index. When $a_k > 0$, completing a square gives

$$G_k = \frac{a_k}{2s}\left\| x_k - z + \frac{1-a_k}{a_k} v_k \right\|^2 - \frac{K_k}{a_k}. \tag{9}$$

Since $K_k \le Q_k^2$ and $a_k = Q_k/Q_0$, it follows that

$$\Psi_k \ge 2Q_0 Q_k + E_k \ge 0. \tag{10}$$

At a zero-energy index, $a_k = K_k = p_k = 0$, so both Ψ_k and the right-hand side vanish directly.

For the upper bound, Cauchy–Schwarz and (7) imply

$$|p_k| \le R\sqrt{\frac{2}{s}}\, Q_k, \qquad a_k h_k \le \frac{R^2}{2sQ_0} Q_k, \qquad E_k \le Q_0 Q_k.$$

Dropping the nonpositive term $-(2-a_k)K_k$ in (8) gives

$$0 \le \Psi_k \le \left(\frac{R^2}{2sQ_0} + R\sqrt{\frac{2}{s}} + 4Q_0 \right) Q_k \le C Q_k. \tag{11}$$

In particular, $\Psi_0 \le C Q_0 \le C^2$, since $C \ge 7Q_0$.

The decisive property is

$$\Psi_{k+1} \le \Psi_k - E_k. \tag{12}$$

To prove it, first note the exact identity

$$h_{k+1} - h_k = p_{k+1} - K_{k+1}. \tag{13}$$

Use (4) with $u = z$, expand $\|y_k - z\|^2$, and apply (13). The result is

$$E_{k+1} \le (1-a_k)p_k - p_{k+1} + (1-a_k)^2 K_k + 2K_{k+1}. \tag{14}$$

Abbreviate $a = a_k$ and $b = a_{k+1}$. Direct expansion of (8), using (13), gives

$$\begin{aligned} G_{k+1} - G_k &= p_{k+1} - (1-a)p_k + (b-a)h_k - 2K_{k+1} + (2-a)K_k \\ &\le (3 - 3a + a^2)K_k - E_{k+1} + (b-a)h_k. \end{aligned} \tag{15}$$

For $Q_k > 0$, the energy decrease (6) also gives

$$Q_0(Q_k - Q_{k+1}) = Q_0 \frac{E_k - E_{k+1}}{Q_k + Q_{k+1}} \ge \frac{2-a}{2} K_k. \tag{16}$$

Combining (15) and (16) yields

$$\Phi_{k+1} - \Phi_k \le -a\left(\frac{3}{2} - a\right) K_k - E_{k+1} + (b-a)h_k \le -E_{k+1}. \tag{17}$$

Here $0 \le b \le a \le 1$ and $h_k \ge 0$. Adding $E_{k+1} - E_k$ proves (12). At $Q_k = 0$, the same conclusion holds by stationarity.

2.3 Extraction of the rate

Proof of Theorem 1. By (11), $E_k \geq \Psi_k^2/C^2$. Thus (12) implies

$$\Psi_{k+1} \leq \Psi_k - \frac{\Psi_k^2}{C^2}.$$

If $\Psi_{k+1} > 0$, monotonicity and positivity give

$$\frac{1}{\Psi_{k+1}} - \frac{1}{\Psi_k} = \frac{\Psi_k - \Psi_{k+1}}{\Psi_k \Psi_{k+1}} \geq \frac{\Psi_k}{C^2 \Psi_{k+1}} \geq \frac{1}{C^2}.$$

Since $\Psi_0 \leq C^2$, summation proves

$$\Psi_m \leq \frac{C^2}{m+1}, \qquad m \geq 0. \tag{18}$$

The bound remains true after a zero of Ψ, because (10) then forces zero energy and stationarity.

Fix $k \geq 2$ and set $m = \lfloor k/2 \rfloor$. Sum (12) from m through $k-1$. Monotonicity of E_k, positivity of Ψ_k, and (18) yield

$$(k-m)E_k \leq \sum_{j=m}^{k-1} E_j \leq \Psi_m - \Psi_k \leq \frac{C^2}{m+1}.$$

For either parity of k, $(m+1)(k-m) \geq k^2/4$. Consequently $E_k \leq 4C^2/k^2$. The two other conclusions follow from $0 \leq F_k \leq E_k$ and $\|v_k\|^2 = 2sK_k \leq 2sE_k$. ∎

3 A discrete bound without a radius assumption

The auxiliary energy in Section 2 is positive and decreases before the sublevel radius is used. Retaining its damping-drop term gives a logarithmic replacement for that radius.

Theorem 2. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and continuously differentiable with L-Lipschitz gradient, $L > 0$, and nonempty minimizer set S. Use (1) with $x_{-1} = x_0$, $E_0 > 0$, and any fixed $0 < s \leq 1/L$. No boundedness of the orbit, sublevels, or distance to S is assumed. Fix any $z \in S$, and put*

$$B = \sqrt{\frac{\|x_0 - z\|^2}{2sE_0} + 4}. \tag{19}$$

Then every integer $n \geq 1$ satisfies

$$E_n \leq \frac{16E_0[B + 8\log n]^4}{n^2}. \tag{20}$$

In particular, for every $\delta > 0$,

$$E_n = o(n^{-2+\delta}), \qquad f(x_n) - f_\star = o(n^{-2+\delta}).$$

The constant uses only the initial data and a chosen minimizer. The estimate does not establish the endpoint $E_n = O(n^{-2})$ without a radius assumption, nor does it assert that the logarithmic loss is necessary. The separate zero-initial-energy case remains stationary.

Proof. Keep h_k, p_k, K_k and Ψ_k from (5) and (8). Their definitions require no radius bound. Equations (10) and (17), before its last simplification, give

$$\begin{aligned} \Psi_k &\geq 2Q_0Q_k + E_k \geq 0, \\ \Psi_{k+1} &\leq \Psi_k - E_k - a_k\left(\frac{3}{2} - a_k\right)K_k - (a_k - a_{k+1})h_k. \end{aligned} \tag{21}$$

Finite zero energy has already been shown to imply stationarity. All divisions below are restricted to the positive-energy part of the orbit.

Put

$$w_k = \frac{1-a_k}{a_k}v_k, \qquad e_k = \frac{K_k}{a_k^2}, \qquad H_k = \frac{\Psi_k}{a_k}, \qquad J_k = \sqrt{H_k}.$$

The square completion (9) becomes

$$H_k = \frac{\|x_k - z + w_k\|^2}{2s} - e_k + 3E_0 + a_kE_0. \tag{22}$$

Since $0 \leq e_k \leq E_0$ and $0 < a_k \leq 1$, this implies

$$\frac{\|w_k\|}{\sqrt{2s}} \leq Q_0, \qquad \sqrt{h_k} \leq J_k + Q_0, \qquad J_k \geq \sqrt{2}Q_0. \tag{23}$$

Using the original expression for H_k and discarding its nonpositive kinetic term gives

$$\begin{aligned} H_k - h_k &\leq 2Q_0\sqrt{h_k} + 4E_0 \\ &\leq 2Q_0J_k + 6E_0 \leq (2+3\sqrt{2})Q_0J_k. \end{aligned} \tag{24}$$

Write $a = a_k$, $b = a_{k+1} > 0$, and $d = a - b \geq 0$. Dividing the second inequality in (21) by b, and dropping its remaining nonpositive kinetic term, gives

$$H_{k+1} \leq H_k + \frac{d}{b}(H_k - h_k) - \frac{E_0a^2}{b}. \tag{25}$$

For $1 \leq a/b \leq 2$, concavity of the square root and (24) imply

$$J_{k+1} - J_k \leq \frac{2+3\sqrt{2}}{2}Q_0\left(\frac{a}{b} - 1\right) \leq 8Q_0\log\frac{a}{b}.$$

Here $r - 1 \leq 2\log r$ for $1 \leq r \leq 2$, and $2 + 3\sqrt{2} < 8$.

For $a/b > 2$, use the position update directly. The energy bounds give

$$\frac{\|v_{k+1}\|}{\sqrt{2s}} \leq Q_0b, \qquad \frac{\|w_{k+1}\|}{\sqrt{2s}} \leq Q_0.$$

Equations (22)–(23) therefore give

$$J_{k+1}^2 \leq [\sqrt{h_k} + Q_0b + Q_0]^2 + 4E_0 \leq [J_k + 3Q_0]^2 + 4E_0.$$

It follows that

$$J_{k+1} \le J_k + 5Q_0 \le J_k + 8Q_0 \log(a/b),$$

using $5 < 8 \log 2$. Thus both relative-drop ranges have the same logarithmic bound. Since $a_0 = 1$ and $v_0 = 0$, summation yields

$$J_k \le \sqrt{h_0 + 4E_0} + 8Q_0 \log(1/a_k) = Q_0[B + 8 \log(1/a_k)]. \tag{26}$$

In particular,

$$\Psi_k = a_k J_k^2 \le E_0 a_k [B + 8 \log(1/a_k)]^2. \tag{27}$$

This comparison uses no bound on the actual position.

Fix $n \ge 1$ with $a_n > 0$, and write $D_n = B + 8 \log(1/a_n)$. For each integer $\ell \ge 0$, collect the indices $j \in \{0, \dots, n-1\}$ satisfying

$$2^\ell a_n \le a_j < 2^{\ell+1} a_n.$$

Every nonempty collection is a consecutive block by monotonicity of a_j. If its first index is m and its size is N_ℓ, telescope (12) across the block and use positivity at the last endpoint. Equation (27) gives

$$N_\ell E_0 \, 2^{2\ell} a_n^2 \le \sum_{\text{block}} E_j \le \Psi_m \le E_0 \, 2^{\ell+1} a_n D_n^2.$$

The blocks partition the n indices. Summing the resulting bounds for N_ℓ gives

$$n a_n \le 4[B + 8 \log(1/a_n)]^2. \tag{28}$$

If $a_n \ge 1/n$, the logarithm on the right is at most $\log n$. If $a_n < 1/n$, the next inequality is automatic, since $B \ge 2$. Thus in either case

$$a_n \le \frac{4[B + 8 \log n]^2}{n}.$$

Squaring and using $E_n = E_0 a_n^2$ proves (20). After a finite zero, the same bound follows by stationarity. Finally, every positive power of n dominates $(\log n)^4$, proving both stated little-o consequences. ∎

4 An interval of limit points

The following example uses the flat-bottom quadratic family tested in [1, Section 5.2]. Its role here is to separate energy decay from convergence of the iterates, and to give lower subsequences for the $O(k^{-2})$ energy and objective bounds.

Proposition 4.1. Let

$$f(x) = \frac{1}{2}\big(\max\{|x| - 1, 0\}\big)^2, \qquad s = \frac{1}{2}, \qquad x_{-1} = x_0 = 2. \tag{29}$$

Then f is convex, continuously differentiable, has globally 1-Lipschitz derivative and bounded sublevels, and $\operatorname{argmin} f = [-1, 1]$. The sequence generated by (1) has cluster set exactly $[-1, 1]$. In particular,

$$\liminf_{k\to\infty} x_k = -1, \qquad \limsup_{k\to\infty} x_k = 1. \tag{30}$$

Its energy is positive at every finite index and tends to zero. Moreover,

$$\frac{1}{256} \le \limsup_{k\to\infty} k^2 E_k < \infty, \qquad \frac{1}{65536} \le \limsup_{k\to\infty} k^2 f(x_k) < \infty. \tag{31}$$

Thus neither energy nor objective is $o(k^{-2})$ on this fixed orbit.

Proof. The derivative is $f'(x) = x - P_{[-1,1]}(x)$, which is nondecreasing and 1-Lipschitz. Every nonempty sublevel is a bounded interval. Define the gradient-step map

$$T(y) = y - \frac{1}{2}f'(y) = \begin{cases} (y-1)/2, & y < -1, \\ y, & |y| \le 1, \\ (y+1)/2, & y > 1. \end{cases}$$

It is strictly increasing and satisfies

$$T(y) \in [-1, 1] \quad \Longleftrightarrow \quad y \in [-1, 1], \qquad T(y) = y \quad \text{on } [-1, 1]. \tag{32}$$

Write $\rho_k = 1 - a_k$ for the inertial coefficient. Here

$$E_k = f(x_k) + v_k^2, \qquad a_k = \sqrt{2E_k}.$$

The first update is exactly

$$x_1 = \frac{3}{2}, \qquad v_1 = -\frac{1}{2}, \qquad E_1 = \frac{3}{8} < E_0 = \frac{1}{2}. \tag{33}$$

By (7), $0 < \rho_k \le 1$ for $k \ge 1$. If $E_{k+1} = 0$, then $x_{k+1} = x_k \in [-1, 1]$ and $T(y_k) = x_k$. Property (32) forces $y_k = x_k$, hence $\rho_k v_k = 0$. For $k \ge 1$ this implies $v_k = 0$ and $E_k = 0$, excluding a first zero. Equation (33) treats the initial update. Therefore $E_k > 0$ at every finite index.

Theorem 1 applies, since the initial sublevel is $[-2, 2]$. It gives

$$E_k \longrightarrow 0, \quad v_k \longrightarrow 0, \quad \operatorname{dist}(x_k, [-1, 1]) \longrightarrow 0, \quad \rho_k \longrightarrow 1. \tag{34}$$

It also supplies the finite upper limsups in (31).

Flat passages. If two successive positions lie in $[-1, 1]$, then (32) gives $v_{k+1} = \rho_k v_k$. The velocity is nonzero there, since zero velocity at a minimizer would mean zero energy.

Its sign is consequently preserved while the positions stay in the interval. With $u_k = |v_k|$, the exact recursion is

$$u_{k+1} = u_k(1 - \sqrt{2}u_k), \qquad \frac{1}{u_{k+1}} - \frac{1}{u_k} = \frac{\sqrt{2}}{1 - \sqrt{2}u_k}. \tag{35}$$

If the sequence stayed in the interval forever after an index $j \geq 1$, then u_k would decrease and the last denominator would be bounded below by $1 - \sqrt{2}u_j > 0$. Thus u_k would be bounded below by the reciprocal of an affine function of k. Its sum would diverge, contradicting bounded travel with a fixed sign. Every such passage therefore exits in finite time, through the endpoint in the direction of its velocity.

Exterior excursions. Choose a tail on which $\rho_k \geq 3/4$ and $|v_k| \leq 1/4$. On the right, write $w_k = x_k - 1 > 0$. If the next point is still on the right, then $y_k > 1$, and the exact updates are

$$w_{k+1} = \frac{w_k + \rho_k v_k}{2}, \qquad v_{k+1} = \frac{\rho_k v_k - w_k}{2}. \tag{36}$$

An exterior run cannot persist forever. Indeed, if $v_k \geq 0$, set $r = v_k/w_k$. As long as the next velocity is also nonnegative, its ratio is $(\rho_k r - 1)/(\rho_k r + 1) < 1$. The following velocity is then negative. Once $v_k < 0$, set $t = -v_k/w_k > 0$. Persistence on the right requires $\rho_k t < 1$ and gives

$$t_{k+1} = \frac{1 + \rho_k t}{1 - \rho_k t} > 1.$$

If a further update still stays on the right, its next ratio is at least 7, using $\rho \geq 3/4$. The positivity condition for the following update is then impossible. Thus every late right excursion exits after finitely many steps; reflection gives the same conclusion on the left.

A right-excursion exit has negative velocity and, since $|v_k| \leq 1/4$, lands in $(3/4, 1]$. A left-excursion exit lands in $[-1, -3/4)$ with positive velocity. Combining finite exterior exits with the flat-passage argument gives infinitely many alternating strict visits to $x > 1$ and $x < -1$. Exact contacts at ± 1 lie in the flat interval; their nonzero directed velocities are covered by (35). The distance limit in (34) now gives (30). Every interior point is crossed on infinitely many passages, and $v_k \to 0$ makes the adjacent crossing positions approach that point. All cluster points lie in $[-1, 1]$ by (34). This proves the full cluster-set assertion.

Lower rate subsequences. Consider a sufficiently late passage from the right exterior to the left exterior. Let j be its first interior index and q its last interior index, just before $x_{q+1} < -1$. Put $m = q - j$ and $u_k = -v_k > 0$ along this passage. The tail bounds imply $x_j > 3/4$ and $x_q < -3/4$, so

$$1 \leq x_j - x_q = \sum_{k=j+1}^{q} u_k \leq m u_j. \tag{37}$$

For $j \leq k < q$, (35) and $\rho_k \geq 3/4$ yield

$$\frac{1}{u_{k+1}} - \frac{1}{u_k} = \frac{\sqrt{2}}{\rho_k} < 2.$$

Hence $1/u_q \le 1/u_j + 2m$. Since $mu_j \ge 1$ by (37),

$$qu_q \ge mu_q \ge \frac{mu_j}{1+2mu_j} \ge \frac{1}{3}, \qquad q^2E_q = q^2u_q^2 \ge \frac{1}{9}. \tag{38}$$

The first exterior overshoot may be tiny, so use two updates. Put $d = x_q + 1 \ge 0$ and $u = u_q$. The strict left exit means $d < \rho_q u$. Direct evaluation of T gives

$$x_{q+1} = -1 - \delta, \quad \delta = \frac{\rho_q u - d}{2} > 0, \quad -v_{q+1} = \frac{\rho_q u + d}{2} \ge \frac{\rho_q u}{2}.$$

The next extrapolated point is still strictly left of the interval. Consequently

$$x_{q+2} = -1 - \delta_2, \qquad \delta_2 = \frac{\delta + \rho_{q+1}(-v_{q+1})}{2} \ge \frac{\rho_{q+1}\rho_q u}{4} \ge \frac{9u}{64}.$$

Equations (38) and $q+2 \ge q$ give

$$(q+2)^2 f(x_{q+2}) \ge \frac{81}{8192} q^2u_q^2 \ge \frac{9}{8192}.$$

Infinitely many passages produce indices $q \to \infty$. These estimates imply both conservative lower bounds in (31). The formulas also cover $x_q = -1$, for which $d = 0$, so no boundary-contact exception is used. ∎

5 The continuous energy bound

The continuous damping law is unnormalized: the friction coefficient is $\sqrt{E(t)}$, without division by its initial value. In this section, the existence of a bounded global classical solution is an explicit hypothesis.

Theorem 3. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and continuously differentiable, with $S = \operatorname{argmin} f \neq \varnothing$ and $f_\star = \min f$. Suppose that $x \in C^2([0,\infty);\mathbb{R}^d)$ is a bounded global solution of*

$$\ddot{x} + \sqrt{E}\,\dot{x} + \nabla f(x) = 0, \qquad E = f(x) - f_\star + \frac{1}{2}\|\dot{x}\|^2. \tag{39}$$

For any $z \in S$, put

$$R = \sup_{t\ge 0} \|x(t) - z\|, \qquad B = \frac{R^2}{2} + \sqrt{2}R + 3. \tag{40}$$

Then, for every $t > 0$,

$$E(t) \le \frac{4B^2}{t^2}, \qquad f(x(t)) - f_\star \le \frac{4B^2}{t^2}, \qquad \|\dot{x}(t)\| \le \frac{2\sqrt{2}B}{t}. \tag{41}$$

If $E(0) = 0$, the given solution is stationary.

The radius in (40) bounds the actual trajectory. No estimate of that radius from initial data, automatic boundedness, or point convergence is asserted. The statement does not require a globally Lipschitz gradient; its assumed classical solution supplies the regularity needed below.

Proof. Write $F = f(x) - f_\star$, $K = \|\dot{x}\|^2/2$, and $Q = \sqrt{E}$. The classical chain rule gives

$$E' = -2QK \leq 0. \tag{42}$$

If $E(0) = 0$, then E and K vanish identically, which proves stationarity without any uniqueness assumption. Suppose henceforth that $E(0) > 0$. On every interval of positive energy,

$$Q' = -K \geq -Q^2, \qquad (1/Q)' = K/Q^2 \leq 1.$$

Thus $Q(t) \geq (1/Q(0) + t)^{-1}$ up to any putative first zero, and continuity excludes such a zero at finite time. In particular, all divisions by Q below are legitimate.

Set $h = \|x - z\|^2/2$ and $p = \langle x - z, \dot{x}\rangle$. Convexity yields

$$h' = p, \qquad p' = 2K - Qp - \langle x - z, \nabla f(x)\rangle \leq 2K - Qp - F.$$

Define

$$\Phi = p + Qh + 3Q. \tag{43}$$

Its derivative satisfies

$$\Phi' \leq 2K - Qp - F - Kh + Qp - 3K = -E - Kh \leq -E. \tag{44}$$

Moreover,

$$p + Qh = \frac{Q}{2}\left\|x - z + \frac{\dot{x}}{Q}\right\|^2 - \frac{K}{Q}, \qquad K \leq Q^2,$$

so $\Phi \geq 2Q > 0$. Cauchy–Schwarz and the actual radius bound give

$$2Q \leq \Phi \leq \left(\sqrt{2}R + \frac{R^2}{2} + 3\right) Q = BQ. \tag{45}$$

Consequently $\Phi' \leq -\Phi^2/B^2$. Integrating its reciprocal gives

$$\Phi(t) \leq \frac{B^2}{t + B^2/\Phi(0)} \leq \frac{B^2}{t}, \qquad t > 0.$$

For any $t > 0$, monotonicity of the true energy and (44) imply

$$\frac{t}{2}E(t) \leq \int_{t/2}^{t} E(u)\,du \leq \Phi(t/2) - \Phi(t) \leq \frac{2B^2}{t}.$$

This proves the full-energy estimate; its nonnegative components give the remaining two bounds. ∎

6 A continuous bound without a radius assumption

The logarithmic loss also removes the radius assumption for the continuous law. The statement concerns a given global classical solution; it does not assert existence or boundedness of that solution.

Theorem 4. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and continuously differentiable, with nonempty minimizer set S and minimum $f_\star$. Let $x \in C^2([0,\infty);\mathbb{R}^d)$ be any global solution of (39). No boundedness of the solution, its distance to S, or any sublevel is assumed. Suppose $E(0) > 0$, put $Q_0 = \sqrt{E(0)}$, and fix any $z \in S$. Define*

$$h_0 = \frac{1}{2}\|x(0) - z\|^2, \qquad p_0 = \langle x(0) - z, \dot{x}(0)\rangle, \qquad W_0 = \sqrt{h_0 + \frac{p_0}{Q_0} + 3}. \tag{46}$$

The expression under the square root is at least 2. For every $t > 0$,

$$E(t) \le \frac{4[W_0 + 3\log(1 + Q_0 t)]^4}{t^2}. \tag{47}$$

Consequently, for every $\delta > 0$,

$$E(t) = o(t^{-2+\delta}), \qquad f(x(t)) - f_\star = o(t^{-2+\delta}).$$

If $E(0) = 0$, the given solution is stationary.

All logarithms are natural. The estimate gives $O(\log^4(2+t)/(1+t)^2)$ as $t \to \infty$, with an initial-data constant, but it does not remove the logarithm from the endpoint estimate. No optimality is asserted for its coefficient or logarithmic power.

Proof. Write $v = \dot{x}$, $F = f(x) - f_\star$, and $Q = \sqrt{E}$. The chain rule gives $E' = -Q\|v\|^2$. Zero initial energy therefore forces stationarity without a uniqueness assumption. If $E(0) > 0$, then on positive-energy intervals

$$Q' = -\frac{1}{2}\|v\|^2 \ge -Q^2, \qquad Q(t) \ge \frac{Q_0}{1 + Q_0 t}.$$

As in the proof of Theorem 3, this lower bound excludes a finite first zero. Thus all divisions below are justified for every finite time.

Set

$$h = \frac{1}{2}\|x - z\|^2, \qquad p = \langle x - z, v\rangle, \qquad \sigma = \frac{p}{Q}, \qquad H = h + \sigma + 3, \qquad W = \sqrt{H}.$$

Cauchy–Schwarz and $\|v\| \le \sqrt{2}Q$ imply $|\sigma| \le 2\sqrt{h}$. Hence

$$H \ge (\sqrt{h} - 1)^2 + 2 \ge 2. \tag{48}$$

In particular $W(0) = W_0$. The elementary identity

$$\frac{9}{2}H - (\sigma + 3)^2 = \frac{9}{2}\left(h - \frac{\sigma^2}{4}\right) + \frac{(\sigma - 6)^2}{8} \ge 0$$

shows that $\sigma + 3 \le (3/\sqrt{2})W \le 6W$.

Let $D = \langle x - z, \nabla f(x)\rangle - F \ge 0$ and $\lambda = -Q'/Q \ge 0$. Using $h' = p$, the equation of motion, and $\|v\|^2 = -2Q'$, direct differentiation gives

$$H' = -Q + \lambda(\sigma + 3) - \frac{D}{Q}. \tag{49}$$

Thus

$$H' + Q \le 6\lambda W, \qquad W' \le 3\lambda - \frac{Q}{2W} \le 3\lambda.$$

Define

$$A(t) = W_0 + 3\int_0^t \lambda(u)\,du = W_0 + 3\log\frac{Q_0}{Q(t)}.$$

Integration gives $W(t) \le A(t)$. Since $A' = 3\lambda$ and $A(0)^2 = H(0)$,

$$H' + Q \le 6\lambda A = (A^2)', \qquad \int_0^t Q(u)\,du \le A(t)^2 - H(t) \le A(t)^2.$$

Monotonicity of Q now yields $tQ(t) \le A(t)^2$. The earlier lower bound for Q gives

$$\log\frac{Q_0}{Q(t)} \le \log(1 + Q_0 t).$$

Therefore

$$E(t) = Q(t)^2 \le \frac{[W_0 + 3\log(1 + Q_0 t)]^4}{t^2},$$

which implies the stated conservative bound (47). The two little-o conclusions follow because $t^{-\delta}(\log t)^4 \to 0$ and $0 \le F \le E$. ∎

7 The continuous quadratic endpoint

The continuous law of [1] uses the damping coefficient $\sqrt{E(t)}$, without division by its initial value. To retain that normalization exactly, consider

$$\ddot{x} + \sqrt{\frac{x^2 + \dot{x}^2}{2}}\,\dot{x} + x = 0, \qquad x(0) = 1, \quad \dot{x}(0) = 0, \tag{50}$$

with

$$f(x) = \frac{x^2}{2}, \qquad E(t) = \frac{x(t)^2 + \dot{x}(t)^2}{2}.$$

This is a fixed scalar, smooth, strongly convex objective with a prescribed zero-velocity start.

Proposition 7.1. The solution of (50) exists uniquely for every $t \ge 0$, and both x and $\dot{x}$ are bounded. Moreover,

$$\lim_{t\to\infty} t^2 E(t) = 4, \qquad \limsup_{t\to\infty} t^2 f(x(t)) = 4, \qquad \liminf_{t\to\infty} t^2 f(x(t)) = 0. \tag{51}$$

In particular, $f(x(t)) = O(t^{-2})$ and $f(x(t)) \ne o(t^{-2})$.

Proof. Put $v = \dot{x}$. The vector field

$$(x, v) \longmapsto \left(v, -x - \frac{\sqrt{x^2 + v^2}}{\sqrt{2}} v \right)$$

is locally Lipschitz. Along its unique local solution,

$$\dot{E} = -\sqrt{E}\, v^2 \leq 0. \tag{52}$$

Since $E(0) = 1/2$, the phase state remains in the unit disk. The solution therefore extends globally. It cannot reach $(0, 0)$ at a finite time, since uniqueness applied backward from such a time would contradict the initial state.

Let $Q = \sqrt{E} > 0$ and choose the continuous lifted phase θ, starting at $\theta(0) = 0$, so that

$$x = \sqrt{2} Q \cos\theta, \qquad v = -\sqrt{2} Q \sin\theta.$$

Differentiating gives the exact equations

$$\dot{Q} = -Q^2 \sin^2\theta, \qquad \dot{\theta} = 1 - Q \sin\theta \cos\theta. \tag{53}$$

They hold also at the zeros of x or v, because only the nonzero radius is divided out. Define

$$m = 1 - \frac{1}{2\sqrt{2}}, \qquad M = 1 + \frac{1}{2\sqrt{2}}.$$

The bound $Q \leq Q(0) = 1/\sqrt{2}$ implies

$$0 < m \leq \dot{\theta} \leq M, \qquad mt \leq \theta(t) \leq Mt. \tag{54}$$

Set $q = 1/Q$. Then $\dot{q} = \sin^2\theta$. Changing variable from time to phase and using (54) gives

$$\begin{aligned} q(t) &= \sqrt{2} + \int_0^t \sin^2\theta(u)\, du \\ &\geq \sqrt{2} + \frac{1}{M} \int_0^{\theta(t)} \sin^2 r\, dr \geq A + Bt, \end{aligned}$$

where $A = \sqrt{2} - 1/(4M) > 0$ and $B = m/(2M) > 0$. Hence

$$Q(t) \leq \frac{1}{A + Bt}, \qquad \int_0^t Q(u)\, du \leq \frac{1}{B} \log\left(1 + \frac{Bt}{A}\right). \tag{55}$$

To determine the asymptotic constant, put

$$I(t) = \int_0^t \cos(2\theta(u))\, du.$$

Integration of the derivative of $\sin(2\theta)$ gives the exact identity

$$I(t) = \frac{\sin(2\theta(t))}{2} - \int_0^t (\dot{\theta}(u) - 1) \cos(2\theta(u))\, du.$$

Since $|\dot{\theta} - 1| \le Q/2$, (55) proves

$$|I(t)| \le \frac{1}{2} + \frac{1}{2B} \log\left(1 + \frac{Bt}{A}\right) = o(t).$$

The identity $\sin^2\theta = (1 - \cos(2\theta))/2$ now yields

$$q(t) = \sqrt{2} + \frac{t}{2} - \frac{I(t)}{2}, \qquad \frac{q(t)}{t} \longrightarrow \frac{1}{2}.$$

Thus $tQ(t) \to 2$ and $t^2 E(t) \to 4$.

Finally, $f(x(t)) = E(t)\cos^2\theta(t)$. The phase bounds (54) show that θ increases strictly to infinity. At the unique times with $\theta = j\pi$, the objective equals E; at the unique times with $\theta = \pi/2 + j\pi$, it is zero. Both sequences of times tend to infinity. Together with $0 \le f(x(t)) \le E(t)$, these exact phase crossings prove the two objective limits in (51). ∎

The continuous energy obstruction has a natural precedent in the lower energy estimates of [1, Proposition 4.11 and Remark 4.12]. Proposition 7.1 adds an exact computation for the objective itself. It does not assert a lower rate for the discrete iteration, whose damping is normalized by Q_0, nor does it establish sharpness of the discrete constant in Theorem 1.

References

[1] S. Maier, C. Castera, and P. Ochs, *Near-optimal closed-loop method via Lyapunov damping for convex optimization*, arXiv:2311.10053v2, 15 April 2024.

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